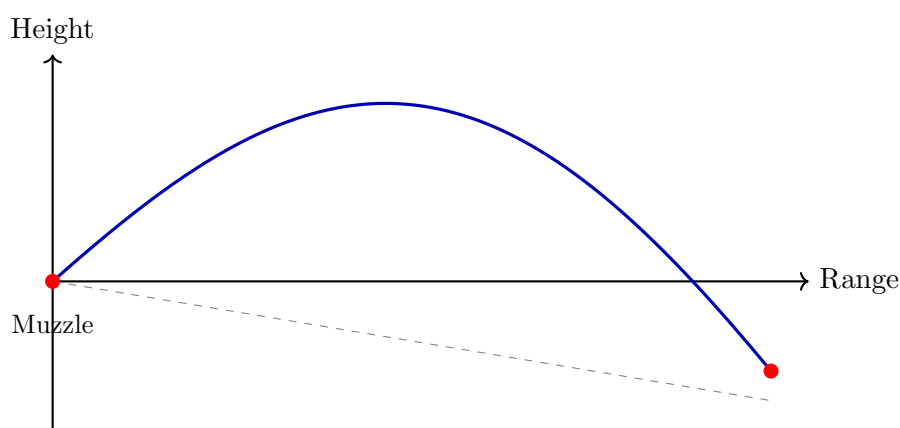


Competition Rifle Ballistics

A Comprehensive Mathematical Manual
with Reloading Guide and Julia Implementations



Trajectory with aerodynamic drag and gravity

*Covering: Interior, Transitional, Exterior, and Terminal Ballistics
Drag Models (G1–G8), Coriolis & Eötvös Effects, Spin Drift,
Aerodynamic Jump, Wind Deflection, and Handloading Data*

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Notation and Conventions

Throughout this manual, the following conventions are used:

Symbol	Meaning	Typical Units
m	Projectile mass	gr or kg
d	Projectile caliber (diameter)	in or m
$v, \mathbf{v}$	Velocity (scalar, vector)	ft/s or $\frac{\text{m}}{\text{s}}$
v_0	Muzzle velocity	ft/s or $\frac{\text{m}}{\text{s}}$
C_D	Drag coefficient (dimensionless)	—
B_C	Ballistic coefficient	$\frac{\text{lbf}}{\text{in}^2}$
SD	Sectional density = m/d^2	$\frac{\text{lbf}}{\text{in}^2}$
i or FF	Form factor (dimensionless)	—
ρ	Air density	$\frac{\text{kg}}{\text{m}^3}$
A	Cross-sectional area = $\pi d^2/4$	m^2
g	Gravitational acceleration	$\frac{\text{m}}{\text{s}^2}$
Ma	Mach number = v/a	—
a	Local speed of sound	$\frac{\text{m}}{\text{s}}$
T	Temperature	K or °F
P	Barometric pressure	hPa or in Hg
H	Relative humidity	%
θ	Launch (elevation) angle	° or mil
ϕ	Latitude	°
ψ	Azimuth (heading)	°
ω_E	Earth's angular velocity	$7.2921 \times 10^{-5} \frac{\text{rad}}{\text{s}}$
S_g	Gyroscopic stability factor	—
τ	Twist rate	in. per turn
r	Range	yd or m
E_k	Kinetic energy	J or ft·lbf

Default values. Parameters shown in [\[default: blue brackets\]](#) indicate default values used when the specific value is unknown. These are based on ICAO standard atmosphere and common competition shooting conditions.

Unit systems. All derivations use SI units internally. Conversion factors to Imperial units (grains, feet per second, inches, etc.) are provided throughout.

Chapter 1

Introduction to Competition Rifle Ballistics

1.1 Scope and Purpose

This manual provides a rigorous mathematical treatment of the ballistic phenomena relevant to precision rifle competition, including but not limited to: F-Class, Palma, Benchrest, PRS (Precision Rifle Series), NRL (National Rifle League), High-Power, and long-range target shooting. The goal is to equip the competitive shooter with both the theoretical foundations and the practical computational tools needed to predict bullet trajectories to the highest possible accuracy.

Competition shooting demands sub-MOA accuracy at distances often exceeding 1000 yd. At these ranges, small errors in modeling atmospheric drag, spin drift, Coriolis deflection, or wind become dominant factors in the miss distance. This manual addresses each of these systematically.

1.2 Overview of Ballistic Phases

The flight of a bullet from ignition to impact is conventionally divided into four phases [McCoy, 1999]:

1. **Interior Ballistics** — The physics of the bullet inside the barrel, from primer ignition through propellant combustion to muzzle exit. Key outputs: muzzle velocity, chamber pressure, barrel time.
2. **Transitional (Intermediate) Ballistics** — The brief phase as the bullet exits the muzzle and the propellant gases dissipate. Muzzle blast interaction, base drag changes, and initial yaw (tip-off) are determined here.
3. **Exterior Ballistics** — The free-flight trajectory from muzzle to target. This is the dominant focus of competition ballistics: aerodynamic drag, gravity drop, wind deflection, spin drift, Coriolis and Eötvös effects, Magnus force, and aerodynamic jump.
4. **Terminal Ballistics** — Bullet behavior at impact. While less critical for paper-target competition, terminal effects matter for steel-target energy requirements and for hunting-oriented competitions.

1.3 Historical Context

The mathematical study of projectile motion dates to Galileo and Newton, but modern exterior ballistics began with the work of Bashforth (1870), who conducted some of the earliest systematic drag measurements [Bashforth, 1870]. The Siacci method (1880s) introduced the concept of the ballistic coefficient and drag functions that remain in use today [Siacci,

1888]. In the 20th century, the US Army Ballistic Research Laboratory (BRL) at Aberdeen Proving Ground produced the definitive drag function tables—G1 through G8—that form the backbone of modern ballistic computation [McCoy, 1999, Ingalls, 1893].

Modern computational ballistics, exemplified by solvers such as those described by Litz [2011] and implemented in software like Bryan Litz’s Applied Ballistics suite, use 3-DOF and 6-DOF point-mass models with empirically measured drag data and atmospheric corrections.

Chapter 2

Atmospheric Model

The atmosphere is the medium through which the bullet travels. Accurate modeling of air density, speed of sound, and viscosity is prerequisite to all exterior ballistic calculations.

2.1 ICAO Standard Atmosphere

The International Civil Aviation Organization (ICAO) standard atmosphere defines baseline conditions at sea level [ICAO, 1993]:

$$T_0 = 288.15 \text{ K} \quad (59^\circ \text{F}) \quad (2.1)$$

$$P_0 = 101\,325 \text{ Pa} \quad (29.92 \text{ in Hg}) \quad (2.2)$$

$$\rho_0 = 1.2250 \frac{\text{kg}}{\text{m}^3} \quad (2.3)$$

$$a_0 = 340.294 \frac{\text{m}}{\text{s}} \quad (1116.45 \text{ ft/s}) \quad (2.4)$$

$$H_0 = 0\% \quad (\text{dry air}) \quad (2.5)$$

Default values: $T_0 = 288.15 \text{ K}$, $P_0 = 101325 \text{ Pa}$, $\rho_0 = 1.225 \text{ kg/m}^3$, $H_0 = 0\%$.

2.1.1 Temperature Lapse Rate

Below 11 000 m (troposphere), the standard lapse rate is:

$$T(h) = T_0 - L \cdot h, \quad L = 0.0065 \frac{\text{K}}{\text{m}} \quad (2.6)$$

where h is the altitude above sea level in meters.

2.1.2 Pressure–Altitude Relationship

Using the barometric formula for the troposphere:

$$P(h) = P_0 \left(\frac{T(h)}{T_0} \right)^{g_0/(L R_{\text{air}})} = P_0 \left(1 - \frac{L h}{T_0} \right)^{5.2559} \quad (2.7)$$

where $g_0 = 9.806\,65 \frac{\text{m}}{\text{s}^2}$ and $R_{\text{air}} = 287.058 \frac{\text{J}}{\text{kg} \cdot \text{K}}$.

2.2 Air Density

The ideal gas law gives the dry-air density:

$$\rho_{\text{dry}} = \frac{P}{R_{\text{air}} T} \quad (2.8)$$

2.2.1 Humidity Correction

Humid air is *less dense* than dry air because water vapor ($M_w = 18.015 \frac{\text{g}}{\text{mol}}$) is lighter than diatomic nitrogen ($M_{N_2} = 28.014 \frac{\text{g}}{\text{mol}}$) and oxygen ($M_{O_2} = 31.998 \frac{\text{g}}{\text{mol}}$).

The saturation vapor pressure (August–Roche–Magnus formula):

$$e_s(T) = 6.1078 \times 10^{\frac{7.5(T-273.15)}{T-35.85}} \quad [\text{hPa}] \quad (2.9)$$

The partial pressure of water vapor:

$$e = \frac{H}{100} e_s(T) \quad (2.10)$$

Corrected density accounting for humidity (virtual temperature method):

$$\rho = \frac{P - 0.3783 e}{R_{\text{air}} T} \quad (2.11)$$

The factor 0.3783 arises from $(1 - M_w/M_{\text{air}}) = 1 - 18.015/28.964 \approx 0.3780$.

2.3 Speed of Sound

The speed of sound in air depends primarily on temperature:

$$a = \sqrt{\gamma R_{\text{air}} T} = 20.0468 \sqrt{T} \quad (2.12)$$

where $\gamma = 1.4$ is the ratio of specific heats for air.

At the ICAO standard temperature: $a_0 = 20.0468 \sqrt{288.15} = 340.29 \frac{\text{m}}{\text{s}}$.

2.4 Density Altitude

Shooters often use density altitude (DA) as a single parameter that encapsulates the combined effects of temperature, pressure, and humidity on air density. DA is the altitude in the standard atmosphere that has the same density as the actual conditions:

$$\text{DA} = \frac{T_0}{L} \left[1 - \left(\frac{\rho}{\rho_0} \right)^{1/(n-1)} \right] \quad (2.13)$$

where $n = g_0/(L R_{\text{air}}) + 1 = 6.2559$.

A practical approximation widely used in the field [Litz, 2011]:

$$\text{DA} \approx 145442.16 \left(1 - \left(\frac{\rho}{\rho_0} \right)^{0.234969} \right) \quad [\text{feet}] \quad (2.14)$$

2.5 Julia Implementation: Atmospheric Model

```

1 # See CompetitionBallistics.jl, module Atmosphere
2 # Key functions:
3 #   std_temperature(h)           -> T [K]
4 #   std_pressure(h)             -> P [Pa]
5 #   air_density(; P, T, H)      -> rho [kg/m^3]
```

```
6 # speed_of_sound(T)          -> a [m/s]
7 # density_altitude(rho)      -> DA [m]
8 # saturation_vapor_pressure(T) -> e_s [Pa]
```

Listing 2.1: Atmospheric model in Julia

Chapter 3

Interior Ballistics

Interior ballistics governs the conversion of chemical energy stored in propellant into kinetic energy of the projectile. For the competition shooter, the key outputs are muzzle velocity and its consistency (extreme spread and standard deviation).

3.1 The Thermodynamic Problem

3.1.1 Nobel–Abel Equation of State

Propellant gases at high pressure deviate significantly from ideal-gas behavior. The Nobel–Abel equation accounts for the finite co-volume of gas molecules [McCoy, 1999, Carlucci and Jacobson, 2007]:

$$P(V - \eta) = n R_u T_g \quad (3.1)$$

where P is the chamber pressure, V is the gas volume, $\eta = nb$ is the total co-volume, n is the number of moles of gas, $R_u = 8.3145 \frac{\text{J}}{\text{mol}\cdot\text{K}}$ is the universal gas constant, and T_g is the gas temperature.

3.1.2 Energy Balance

The first law of thermodynamics applied to the propellant–projectile system:

$$Q_{\text{prop}} = E_k + E_{\text{heat}} + E_{\text{friction}} + E_{\text{gas KE}} + E_{\text{residual}} \quad (3.2)$$

The total chemical energy released by the propellant charge:

$$Q_{\text{prop}} = C \cdot f \cdot z \quad (3.3)$$

where C is the charge mass, f is the impetus (force constant, energy per unit mass), and z is the fraction of charge burned ($0 \leq z \leq 1$).

3.2 Simplified Closed-Bomb Model

In a closed bomb (no projectile motion), the pressure is:

$$P = \frac{C f z}{V_{\text{ch}} - C(1 - z)/\rho_p - C z \eta_{\text{sp}}} \quad (3.4)$$

where V_{ch} is the chamber volume, ρ_p is the solid propellant density, and η_{sp} is the specific co-volume.

3.3 Burn Rate Law

The rate of gas generation follows a geometric burn rate law. For a single-perforated cylindrical grain:

$$\frac{dz}{dt} = \frac{S(z)}{V_g(z)} \beta P^n \quad (3.5)$$

For small arms propellants, the empirical Vieille law is commonly used:

$$r_b = \beta P^n \quad (3.6)$$

where r_b is the linear regression rate ($\frac{\text{m}}{\text{s}}$), and typical values are $n \approx 0.6\text{--}0.9$.

3.3.1 Geometric Form Function

For common grain geometries, the fraction burned z relates to the linear fraction burned $\xi = e/e_0$ through a form function:

$$z(\xi) = \xi (1 + \theta \xi) \quad (3.7)$$

where θ is the form factor: $\theta = 0$ for a slab, $\theta > 0$ for progressive, and $\theta < 0$ for degressive grains.

3.4 Le Duc Velocity Model

A classical approximation for muzzle velocity useful for handloading estimates [McCoy, 1999]:

$$v(x) = \frac{a x}{b + x} \quad (3.8)$$

where x is the projectile travel distance from the cartridge case mouth, and a and b are empirically determined constants. As $x \rightarrow \infty$, $v \rightarrow a$, so a represents the theoretical maximum velocity.

3.5 Pressure–Travel Relationship

The pressure as a function of bullet travel x in the barrel:

$$P(x) = \frac{m_e}{A_{\text{bore}}} \frac{dv}{dt} = \frac{m_e v}{A_{\text{bore}}} \frac{dv}{dx} \quad (3.9)$$

where $m_e = m + C/3$ is the effective mass and $A_{\text{bore}} = \pi d^2/4$ is the bore cross-sectional area.

Using the Le Duc model, the pressure becomes:

$$P(x) = \frac{m_e a^2 b x}{A_{\text{bore}} (b + x)^3} \quad (3.10)$$

The peak pressure occurs at $x_{\text{max}} = b/2$:

$$P_{\text{max}} = \frac{4 m_e a^2}{27 A_{\text{bore}} b} \quad (3.11)$$

3.6 Muzzle Velocity Estimation for Handloaders

A practical empirical formula due to Powley and refined by [Litz \[2011\]](#):

$$v_0 \approx v_{\text{ref}} \left(\frac{L_{\text{barrel}}}{L_{\text{ref}}} \right)^{0.25 \text{ to } 0.30} \quad (3.12)$$

A rough rule of thumb: each inch of barrel length change alters muzzle velocity by approximately 20 ft/s to 35 ft/s for typical rifle cartridges [[Litz, 2011](#)].

3.6.1 Temperature Sensitivity of Muzzle Velocity

Propellant burn rate is temperature-dependent. The temperature coefficient:

$$\frac{\Delta v_0}{v_0} = \sigma_T \Delta T \quad (3.13)$$

where σ_T is the temperature sensitivity coefficient, typically 0.5 to 2.0 fps/°F depending on propellant. Modern temperature-stable propellants like Hodgdon Extreme series have $\sigma_T \approx 0.5$ fps/°F [[Litz, 2011](#)].

[default: $\sigma_T = 1.0$ fps/°F]

Chapter 4

Transitional Ballistics

Transitional (or intermediate) ballistics covers the brief period after the bullet exits the muzzle and before it settles into free flight. This phase lasts only a few calibers of travel but establishes critical initial conditions for the exterior trajectory.

4.1 Muzzle Gas Dynamics

When the bullet exits the barrel, the high-pressure propellant gases (10–70 MPa) expand rapidly into the atmosphere. The gas jet initially travels faster than the bullet and forms a complex shock-wave structure [Carlucci and Jacobson, 2007]:

1. **Muzzle blast wave:** A spherical shock radiating outward.
2. **Precursor blast:** From air compressed ahead of the bullet in the barrel.
3. **Bottle shock:** The characteristic diamond-pattern Mach disk structure formed by the underexpanded supersonic gas jet.

4.2 Muzzle Tip-Off and Initial Yaw

As the bullet exits the muzzle, asymmetric gas forces can impart a small angular velocity (tip-off rate). The resulting initial yaw angle α_0 is typically small (0.5–3 °F) for well-made ammunition but is a significant factor in group size.

4.3 Base Drag Transition

Inside the barrel, the bullet base is subject to gas pressure. Upon exit, this pressure rapidly drops to ambient, and base drag increases significantly. The base drag coefficient transitions from a suppressed value to its free-flight value over a distance of approximately 5–20 calibers from the muzzle [McCoy, 1999].

4.4 Crown Effects

The muzzle crown—the precise geometry at the barrel’s end—is critical to accuracy because it determines how symmetrically the gases release around the bullet base.

Chapter 5

Exterior Ballistics: Core Theory

This chapter develops the mathematical framework for predicting the bullet's trajectory in free flight.

5.1 Vacuum Trajectory (Baseline)

In the absence of air resistance, the trajectory is a simple parabola:

$$x(t) = v_0 \cos \theta \cdot t \quad (5.1)$$

$$y(t) = v_0 \sin \theta \cdot t - \frac{1}{2} g t^2 \quad (5.2)$$

The range in vacuum:

$$R_{\text{vac}} = \frac{v_0^2 \sin 2\theta}{g} \quad (5.3)$$

5.2 Aerodynamic Drag

5.2.1 The Drag Equation

The aerodynamic drag force opposes the bullet's motion through the air:

$$F_D = \frac{1}{2} \rho v^2 C_D(Ma) A \quad (5.4)$$

The drag deceleration:

$$a_D = \frac{F_D}{m} = \frac{\rho v^2 C_D A}{2m} \quad (5.5)$$

5.2.2 Drag Coefficient and Mach Number

The drag coefficient C_D is fundamentally a function of Mach number $Ma = v/a$, Reynolds number Re , and projectile geometry. For small-arms bullets, Re effects are secondary, and C_D is treated as a function of Ma alone [McCoy, 1999].

The C_D – Ma curve has characteristic regimes: subsonic ($Ma < 0.8$) with low, slowly varying C_D ; transonic ($0.8 < Ma < 1.2$) with a sharp rise; and supersonic ($Ma > 1.2$) with gradually decreasing C_D .

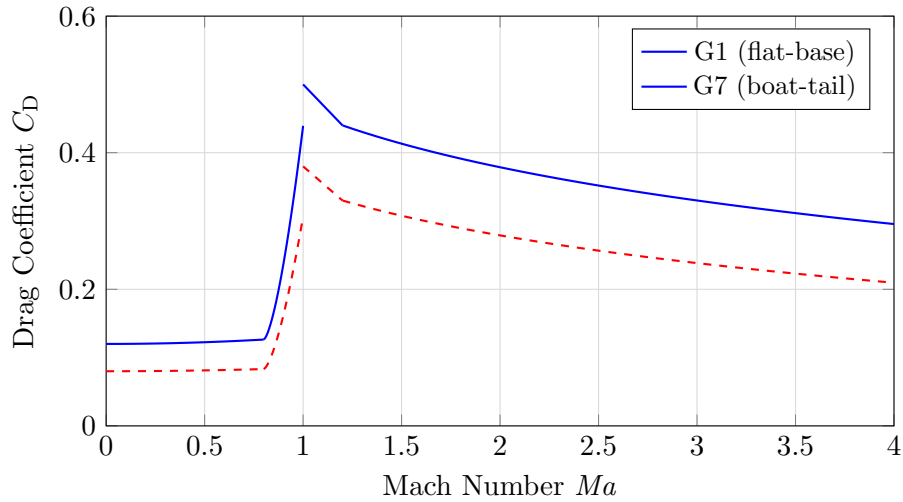


Figure 5.1: Schematic drag coefficient vs. Mach number for G1 and G7 reference projectiles.

5.3 Ballistic Coefficient

The ballistic coefficient B_C is defined as:

$$B_C = \frac{SD}{i} = \frac{m}{i d^2} \quad (5.6)$$

where $SD = m/d^2$ is the sectional density and i is the form factor:

$$i(Ma) = \frac{C_D^{\text{bullet}}(Ma)}{C_D^{\text{std}}(Ma)} \quad (5.7)$$

B_C is only meaningful when referenced to a specific drag model (G1, G7, etc.).

5.3.1 Standard Conditions for BC

By convention, B_C values are quoted at standard atmosphere conditions. To use B_C at non-standard conditions:

$$B_{C\text{actual}} = B_{C\text{std}} \cdot \frac{\rho_0}{\rho_{\text{actual}}} \quad (5.8)$$

5.4 Standard Drag Models (G1 through G8)

Table 5.1: BRL Standard Drag Models [McCoy, 1999, Ingalls, 1893]

Model	Reference Projectile Shape	Primary Use
G1	Flat-base with 2-caliber ogive nose	Traditional default; hunting bullets
G7	Long, secant-ogive, boat-tail	Best for modern VLD match bullets
G8	Similar to G7 with flat base	Flat-base match bullets
GL	Blunt-nose lead bullet	Cast lead bullets

For competition rifle shooting, G7 is strongly preferred over G1 for modern VLD boat-tail bullets because the G7 reference shape more closely matches these bullets, resulting in a form factor i closer to 1.0 and a more constant B_C across the velocity range [Litz, 2011].

5.4.1 G7 Drag Function

Table 5.2: Selected G7 drag coefficient values [McCoy, 1999]

Ma	C_D^{G7}	Ma	C_D^{G7}	Ma	C_D^{G7}
0.00	0.1198	1.00	0.4002	2.00	0.2487
0.50	0.1198	1.05	0.3810	2.50	0.2169
0.80	0.1280	1.10	0.3621	3.00	0.1966
0.90	0.1687	1.20	0.3278	3.50	0.1800
0.95	0.2300	1.50	0.2860	4.00	0.1668

5.5 Equations of Motion: 3-DOF Point-Mass Model

5.5.1 Coordinate System

We use a ground-fixed, flat-Earth coordinate system: x (downrange), y (vertical, upward positive), z (cross-range, positive right).

5.5.2 Vector Equation of Motion

$$m \frac{d\mathbf{v}}{dt} = -\frac{1}{2} \rho \|\mathbf{v}_{\text{rel}}\| \mathbf{v}_{\text{rel}} C_D(Ma) A + m \mathbf{g} \quad (5.9)$$

where $\mathbf{v}_{\text{rel}} = \mathbf{v} - \mathbf{v}_{\text{wind}}$ and $\mathbf{g} = (0, -g, 0)^T$.

5.5.3 Scalar Component Equations

$$\frac{dv_x}{dt} = -\frac{\rho C_D A}{2m} v_{\text{rel}} (v_x - w_x) \quad (5.10)$$

$$\frac{dv_y}{dt} = -\frac{\rho C_D A}{2m} v_{\text{rel}} (v_y - w_y) - g \quad (5.11)$$

$$\frac{dv_z}{dt} = -\frac{\rho C_D A}{2m} v_{\text{rel}} (v_z - w_z) \quad (5.12)$$

where $v_{\text{rel}} = \sqrt{(v_x - w_x)^2 + (v_y - w_y)^2 + (v_z - w_z)^2}$.

5.5.4 Using the Ballistic Coefficient

Substituting $C_D = i \cdot C_D^{\text{std}}$ and $B_C = m/(i d^2)$:

$$\frac{\rho C_D A}{2m} = \frac{\rho C_D^{\text{std}} \pi}{8 B_C} \quad (5.13)$$

5.6 Numerical Integration

The equations of motion are solved numerically using a 4th-order Runge–Kutta (RK4) method [Press et al., 2007]:

$$\mathbf{k}_1 = h \mathbf{f}(t_n, \mathbf{y}_n) \quad (5.14)$$

$$\mathbf{k}_2 = h \mathbf{f}(t_n + h/2, \mathbf{y}_n + \mathbf{k}_1/2) \quad (5.15)$$

$$\mathbf{k}_3 = h \mathbf{f}(t_n + h/2, \mathbf{y}_n + \mathbf{k}_2/2) \quad (5.16)$$

$$\mathbf{k}_4 = h \mathbf{f}(t_n + h, \mathbf{y}_n + \mathbf{k}_3) \quad (5.17)$$

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \frac{1}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4) \quad (5.18)$$

A typical time step of $h = 0.0005$ to 0.001 s is sufficient.

Chapter 6

Exterior Ballistics: Advanced Effects

6.1 Wind Deflection

Wind is the single largest variable in long-range competition shooting.

6.1.1 Wind Components

Given wind speed W at angle α_w relative to the line of fire:

$$w_x = -W \cos \alpha_w \quad (\text{headwind/tailwind}) \quad (6.1)$$

$$w_z = W \sin \alpha_w \quad (\text{crosswind}) \quad (6.2)$$

6.1.2 Lag Rule (Quick Estimation)

The Didion approximation [McCoy, 1999, Litz, 2011]:

$$\Delta z_{\text{wind}} \approx W \sin \alpha_w \cdot \left(t_{\text{actual}} - \frac{R}{v_0 \cos \theta} \right) \quad (6.3)$$

This is remarkably accurate (within 5%) and provides an intuitive physical picture: the wind deflection equals the crosswind speed multiplied by the lag time that drag has cost the bullet.

6.2 Spin Drift

A spin-stabilized bullet develops a slow lateral drift in the direction of its spin. The empirical formula widely used for spin drift [Litz, 2011]:

$$\Delta z_{\text{spin}} = 1.25 (S_g + 1.2) \cdot t^{1.83} \quad (6.4)$$

where Δz_{spin} is the deflection in inches, S_g is the gyroscopic stability factor, and t is the time of flight in seconds.

6.2.1 Gyroscopic Stability Factor

The Miller stability formula [Miller, 2005]:

$$S_g \approx \frac{30 m}{T_w^2 d^3 l (1 + l^2)} \cdot \frac{v_0}{2800} \cdot \frac{T_0}{T} \quad (6.5)$$

where m is bullet mass in grains, T_w is twist rate in calibers per turn, d is caliber in inches, l is bullet length in calibers, $T_0 = 518.67 \text{ R } (59^\circ\text{F})$.

For adequate stability, $S_g > 1.0$; values of 1.3–2.0 are typical for match bullets.

6.3 Coriolis and Eötvös Effects

6.3.1 Coriolis Acceleration

In an Earth-fixed reference frame:

$$\mathbf{a}_{\text{Cor}} = -2\boldsymbol{\omega}_E \times \mathbf{v} \quad (6.6)$$

where $\omega_E = 7.2921 \times 10^{-5} \frac{\text{rad}}{\text{s}}$.

6.3.2 Horizontal Deflection

$$\boxed{\Delta z_{\text{Cor}} \approx \omega_E \sin \phi \cdot R \cdot t} \quad (6.7)$$

Always to the right in the Northern Hemisphere, left in the Southern. At 45 N, 1000 yd, $t \approx 1.5$ s: $\Delta z \approx 2.8$ inches (≈ 0.27 MOA).

6.3.3 Vertical Deflection (Eötvös Effect)

$$\Delta y_{\text{Eot}} \approx -\omega_E \cos \phi \sin \psi \cdot R \cdot t \quad (6.8)$$

6.4 Aerodynamic Jump

For a spin-stabilized bullet with initial yaw α_0 [McCoy, 1999]:

$$\theta_{\text{jump}} = \frac{C_{L\alpha}}{2k_t^2} \frac{\alpha_{\text{trim}}}{p_s - r_s} \quad (6.9)$$

In practice, aerodynamic jump is absorbed into zero-offset and has a random component that contributes to group size rather than a systematic bias.

6.5 Magnus Effect

$$F_{\text{Magnus}} = \frac{1}{2} \rho v^2 A C_{N_{p\alpha}} \frac{p d}{2 v} \alpha \quad (6.10)$$

For typical small yaw angles ($\alpha < 1$ deg), the Magnus contribution is sub-0.1 MOA at 1000 yd and is usually neglected.

Chapter 7

Complete 3-DOF Ballistic Solver

The complete 3-DOF ballistic solver is implemented in the companion file `CompetitionBallistics.jl` (see Appendix B). The solver includes: G1/G7 drag model interpolation from BRL tables, full RK4 integration of the 3-DOF equations, automatic zero-angle finding, Coriolis and Eötvös corrections, post-hoc spin drift, and wind deflection. The `ShotParameters` struct accepts all physical inputs (bullet mass, BC, muzzle velocity, atmospheric conditions, wind, latitude, twist rate, etc.) and the `solve_trajectory()` function returns a vector of `TrajectoryPoint` records containing time, position, velocity, Mach number, and kinetic energy at each integration step.

```
1 include("CompetitionBallistics.jl")
2 using .CompetitionBallistics.ExteriorBallistics
3
4 params = ShotParameters(
5     mass_grains=140.0, caliber_in=0.264,
6     bc=0.311, drag_model=:G7,
7     muzzle_vel_fps=2700.0, zero_range_yd=100.0,
8     wind_speed_mph=10.0, wind_angle_deg=90.0,
9     target_range_yd=1000.0,
10    twist_in=8.0, bullet_length_in=1.34,
11 )
12 trajectory_table(params, step_yd=100)
```

Listing 7.1: Example: running the solver

Chapter 8

Angular Measurement and Sighting Systems

8.1 Minute of Angle (MOA)

$$1 \text{ MOA} = \frac{1}{60} \times \frac{\pi}{180} = 2.9089 \times 10^{-4} \text{ rad} \quad (8.1)$$

The common approximation: 1 MOA $\approx$ 1.047 inches at 100 yd.

8.2 Milliradian (Mil)

$$1 \text{ mil} = 0.001 \text{ rad} \approx 3.6 \text{ inches at } 100 \text{ yd} \quad (8.2)$$

Conversion: 1 mil = 3.438 MOA.

Chapter 9

Reloading Guide for Competition Shooters

Safety warning: Always cross-reference with current published load data. Start 10% below listed maximum charges and work up carefully.

9.1 Cartridge Selection for Competition

Table 9.1: Popular competition cartridges and their characteristics

Cartridge	Bullet (gr)	v_0 (fps)	BC (G7)	Barrel	Discipline
6mm BR	105–108	2850–2950	.225–.275	26–28	Benchrest, F-Class
6.5 Creedmoor	130–147	2550–2800	.255–.340	24–26	PRS, NRL, F-Class
6.5×47 Lapua	130–147	2650–2850	.255–.340	26–28	Palma variant
.284 Win	180	2850–2950	.340–.365	28–30	F-Class Open
.308 Win	155–185	2550–2800	.220–.275	24–30	Palma, Tactical
.223 Rem	69–80	2700–3000	.145–.200	20–26	Service Rifle
.300 Win Mag	190–230	2800–3000	.310–.415	26–30	F-Class Open
.338 Lapua Mag	250–300	2700–2900	.370–.450	27–30	ELR
6mm Dasher	105–115	2950–3050	.225–.290	26–28	PRS, NRL

9.2 Propellant Selection

Table 9.2: Popular propellants for competition rifle cartridges

Propellant	Burn Rate	Temp Stable?	Typical Cartridges
Hodgdon Varget	Medium	Yes (Extreme)	.308, .223, 6.5 CM
Hodgdon H4350	Med-slow	Yes (Extreme)	6.5 CM, .284, .260
Alliant RL-16	Med-slow	Yes (ETS)	6.5 CM, 6mm BR
Vihtavuori N150	Med-slow	Moderate	6.5 CM, .260
Vihtavuori N565	Med-slow	Yes	6.5 CM, 6.5×47
IMR 4451	Med-slow	Yes (Enduron)	6.5 CM, .308

9.3 Representative Load Data

Disclaimer: Reference only. Always consult current manufacturer manuals. Start 10% below maximum.

Table 9.3: 6.5 Creedmoor representative loads (24-inch barrel)

Bullet	Propellant	Charge (gr)	COAL (in)	v_0 (fps)	Press.
140 Berger Hybrid	H4350	40.0–42.5	2.800	2650–2760	Near max
147 Hornady ELD-M	H4350	38.5–41.0	2.830	2550–2660	Near max
130 Berger Hybrid	H4350	41.0–43.5	2.810	2750–2880	Near max
140 Sierra MK	RL-16	39.0–41.5	2.800	2640–2750	Near max
147 Hornady ELD-M	N565	39.5–42.0	2.830	2570–2680	Near max

Table 9.4: .308 Winchester representative loads (24-inch barrel)

Bullet	Propellant	Charge (gr)	COAL (in)	v_0 (fps)	Press.
175 Sierra MK	Varget	42.0–44.5	2.800	2550–2680	Near max
155 Berger Hybrid	Varget	44.5–47.0	2.800	2750–2870	Near max
168 Sierra MK	Varget	43.0–45.5	2.800	2600–2720	Near max
175 Sierra MK	N150	43.0–45.0	2.800	2570–2670	Near max

9.4 Interpreting Chronograph Data

$$\bar{v} = \frac{1}{N} \sum_{i=1}^N v_i, \quad \text{ES} = v_{\max} - v_{\min}, \quad \text{SD} = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (v_i - \bar{v})^2} \quad (9.1)$$

Target values: ES < 20 fps (excellent: < 10), SD < 8 fps (excellent: < 4).

9.4.1 Velocity SD Impact on Vertical Dispersion

$$\sigma_y(R) \approx \left| \frac{\partial y}{\partial v_0} \right|_R \cdot \sigma_v \quad (9.2)$$

Chapter 10

Ballistic Reference Tables

10.1 Bullet Data for Common Competition Bullets

Table 10.1: Ballistic data for popular competition bullets [Litz, 2011, 2015]

Bullet	Wt (gr)	Cal	BC G1	BC G7	SD	Twist
Berger 105 Hyb 6mm	105	.243	0.536	0.275	0.254	1:7.5
Berger 140 Hyb 6.5	140	.264	0.607	0.311	0.287	1:8
Hornady 147 ELD-M	147	.264	0.697	0.351	0.301	1:7.5
Berger 180 Hyb 7mm	180	.284	0.683	0.350	0.319	1:8.5
Sierra 175 MK .308	175	.308	0.505	0.259	0.264	1:10
Berger 185 Jugg .308	185	.308	0.552	0.283	0.279	1:10
Hornady 178 ELD-M	178	.308	0.535	0.274	0.268	1:10
Berger 300 Hyb .338	300	.338	0.818	0.419	0.375	1:9.4

10.2 Sample Trajectory Table

Table 10.2: Trajectory: 6.5 CM, 140 gr Berger Hybrid, $BC_{G7} = 0.311$, $v_0 = 2700$ fps, 100 yd zero, sea level standard atmosphere

Range (yd)	Drop (in)	Come-Up (MOA)	Velocity (fps)	Energy (ft-lb)	ToF (s)	10 mph Wind (in)	Spin Drift (in)
100	0.0	0.0	2545	2015	0.115	0.5	0.0
200	-2.8	1.3	2396	1786	0.237	2.1	0.1
300	-10.7	3.4	2252	1578	0.365	5.0	0.3
400	-24.5	5.9	2113	1389	0.501	9.2	0.5
500	-45.0	8.6	1979	1219	0.646	15.1	0.9
600	-73.4	11.7	1851	1066	0.800	22.8	1.4
700	-111.0	15.1	1728	929	0.965	32.5	2.0
800	-159.5	19.0	1610	807	1.143	44.6	2.8
900	-221.1	23.5	1498	698	1.334	59.3	3.8
1000	-298.0	28.5	1393	604	1.541	77.1	5.1

Chapter 11

Special Topics

11.1 Transonic Stability

The dynamic stability factor S_d must be considered alongside S_g [McCoy, 1999]. For full stability: $S_g > 1.0$ and $0 < S_d < 2$.

11.2 Inclined Fire (Rifleman's Rule)

$$R_{\text{effective}} = R_{\text{slant}} \cdot \cos \alpha \quad (11.1)$$

The perpendicular gravity component $g_{\perp} = g \cos \alpha$ causes the bullet to drop away from the line of sight. Since $\cos \alpha < 1$ for any nonzero inclination (up or down), the drop is always less than flat fire.

11.3 Barrel Harmonics and Optimal Charge Weight

Barrel vibrations cause the muzzle to describe an elliptical pattern at the moment of bullet exit. The OCW method seeks charge weights where the muzzle vertical angle is at an extremum, creating a flat spot that compensates for velocity variation.

Chapter 12

Error Budget and Sensitivity Analysis

12.1 Sensitivity Coefficients

For any trajectory output Y at range R :

$$S_i = \frac{\partial Y}{\partial p_i} \approx \frac{Y(p_i + \Delta p_i) - Y(p_i - \Delta p_i)}{2 \Delta p_i} \quad (12.1)$$

Table 12.1: Error budget: 6.5 Creedmoor, 140 gr, $BC_{G7} = 0.311$, 2700 fps at 1000 yd

Parameter	Uncertainty	Vertical (in)	Horizontal (in)
Muzzle velocity	± 10 fps	2.5	—
BC	$\pm 3\%$	3.2	1.8 (wind)
Wind speed (10 mph cross)	± 2 mph	—	15.4
Wind direction	$\pm 10^\circ$	—	6.8
Temperature	$\pm 10^\circ\text{F}$	0.8	—
Range estimation	± 5 yd	1.5	—
Spin drift (uncompensated)	—	—	5.1
Coriolis (45°N, uncomp.)	—	<1.0	2.8

Key insight: Wind reading dominates the error budget at long range. Wind-related errors are 3–5 times larger than all other sources combined.

Chapter 13

Ballistic Implications for Reloading

This chapter bridges the mathematical models of the preceding chapters with the practical decisions a competition handloader must make at the reloading bench.

13.1 The Reloader's Equation: What You Control

From the 3-DOF equation of motion (Eq. 5.9), the trajectory depends on three groups of quantities:

1. **Bullet parameters** (m, d, B_C, L): fixed once the bullet is selected.
2. **Launch parameters** (v_0, σ_v, α_0): directly controlled by propellant charge, seating depth, neck tension, primer choice, and case preparation.
3. **Environmental parameters** ($\rho, T, P, H, \text{wind}$): not controllable, but the reloader can minimize their *impact* through component choices that reduce trajectory sensitivity.

13.2 Muzzle Velocity: The Central Reloading Output

13.2.1 Charge Weight and Velocity

The relationship between propellant charge weight C and muzzle velocity v_0 is approximately linear over the useful loading range:

$$v_0 \approx v_{\text{base}} + k_C (C - C_{\text{base}}) \quad (13.1)$$

where k_C is the velocity gain per grain, typically 15–40 fps/gr. For 6.5 Creedmoor with H4350, $k_C \approx 25$ fps/gr [Litz, 2011].

13.2.2 Velocity Consistency and Vertical Dispersion

From Eq. (9.2), the vertical group size at range R due to velocity variation:

$$\sigma_y(R) = \left| \frac{\partial y}{\partial v_0} \right|_R \cdot \sigma_v \quad (13.2)$$

Table 13.1: Velocity sensitivity: vertical shift per 1 fps change, 6.5 CM 140 gr

Range (yd)	$ \partial y / \partial v_0 $ (in/fps)	Vertical for SD=10 fps (in)
300	0.02	0.2
600	0.08	0.8
800	0.15	1.5
1000	0.25	2.5
1200	0.40	4.0
1500	0.72	7.2

13.2.3 Achieving Low SD: Quantitative Checklist

Assuming independent error sources, the combined SD is:

$$\sigma_{v,\text{total}} = \sqrt{\sigma_{\text{charge}}^2 + \sigma_{\text{brass}}^2 + \sigma_{\text{primer}}^2 + \sigma_{\text{seat}}^2 + \sigma_{\text{neck}}^2 + \sigma_{\text{lot}}^2} \quad (13.3)$$

Table 13.2: Estimated velocity SD contributions by reloading variable

Variable	Typical σ (fps)	Careful prep (fps)
Charge weight (± 0.1 gr thrown)	6–12	2–4 (weighed ± 0.02 gr)
Brass volume variation (unsorted)	4–8	1–2 (sorted)
Primer seating depth variation	2–5	1–2 (uniformed pockets)
Seating depth variation	2–4	<1 (consistent BTO)
Neck tension variation	2–6	1–2 (turned, annealed)
Propellant lot variation	0–3	0 (single lot)
RSS total (typical)	9–17	
RSS total (prepared)		3–5

13.3 BC Sensitivity and Bullet Selection

A 10% higher BC reduces wind deflection by approximately 8–12%. **Choosing the highest-BC bullet your twist rate can stabilize is the single most impactful choice for reducing wind sensitivity**, which is the dominant error source (Table 12.1).

13.4 Temperature Effects: Propellant and Atmosphere

13.4.1 Combined Temperature Effect

$$\Delta y_{\text{total}}(R, \Delta T) = \underbrace{\frac{\partial y}{\partial v_0} \sigma_T \Delta T}_{\text{velocity effect}} + \underbrace{\frac{\partial y}{\partial \rho} \frac{\partial \rho}{\partial T} \Delta T}_{\text{density effect}} \quad (13.4)$$

For temperature-sensitive propellants, the two effects are *additive*. Using a temperature-stable propellant is ballistically equivalent to reducing the effective sensitivity by 60–80%.

Table 13.3: Temperature sensitivity by propellant class

Propellant Class	σ_T (fps/°F)	Δv_0 for $\pm 40^\circ\text{F}$
Standard (IMR 4064, etc.)	1.5–2.0	± 60 –80 fps
Temperature-stable (Varget, H4350)	0.3–0.8	± 12 –32 fps
Highly stable (RL-16, N565)	0.2–0.5	± 8 –20 fps

13.5 Seating Depth and Internal Ballistic Coupling

Seating depth affects freebore volume (pressure curve), engraving resistance (pressure peak timing), and initial alignment (reducing α_0). The empirical relationship:

$$\frac{\partial v_0}{\partial(\text{jump})} \approx -3 \text{ to } -8 \text{ fps per } 0.001 \text{ in. increase} \quad (13.5)$$

The group size model:

$$G(j) = \sqrt{G_{\text{base}}^2 + G_{\alpha}^2(j) + G_v^2(j)} \quad (13.6)$$

Typical sweet spots: jam/light contact (0.000–0.005 in.), moderate jump (0.015–0.030 in.), and extended jump (0.060–0.100 in.) [Litz, 2011].

13.6 Pressure Estimation and Safety Margins

Table 13.4: SAAMI maximum average pressure for competition cartridges

Cartridge	MAP (psi)	MAP (MPa)
.223 Remington	55,000	379
6.5 Creedmoor	62,000	427
.308 Winchester	62,000	427
.300 Winchester Magnum	64,000	441
.338 Lapua Magnum	60,916	420

From the Le Duc model peak pressure (Eq. 3.11):

$$P_{\text{peak}} \approx 1.6 \times P_{\text{avg}} = 1.6 \times \frac{m_e v_0^2}{2 A_{\text{bore}} L_{\text{travel}}} \quad (13.7)$$

13.7 Barrel Life

Table 13.5: Approximate barrel life for competition cartridges

Cartridge	Barrel Life (rounds)
6mm BR / 6 Dasher	2500–3500
6.5 Creedmoor	2500–3000
.308 Winchester	5000–8000
.284 Winchester	1500–2500
.300 Winchester Magnum	1200–2000

The ballistic signature of barrel wear:

$$v_0(N) \approx v_{0,\text{new}} - k_{\text{wear}} \cdot N^{0.7} \quad (13.8)$$

$$\sigma_v(N) \approx \sigma_{v,\text{new}} + k_{\sigma} \cdot N^{0.5} \quad (13.9)$$

13.8 Load Development Workflow

Based on the mathematical models above:

Step 1: Bullet selection. Maximize B_C . Verify $S_g > 1.4$ using Eq. (6.5).

Step 2: Propellant selection. Temperature-stable. Calculate Δv_0 over competition temperature range using Eq. (3.13).

Step 3: Brass preparation. Sort by weight/capacity. Uniform necks and primer pockets.

Step 4: Charge weight exploration. Satterlee ladder or OCW test. Identify velocity nodes.

Step 5: Seating depth optimization. Test at 0.010 in. increments from lands.

Step 6: Validation. 20+ rounds over chronograph. $SD < 8$ fps (target < 5). Run the solver (Chapter 7) to verify vertical dispersion at max range.

Step 7: Temperature verification. Compute come-up shift using Eq. (13.4).

Appendix A

Detailed Mathematical Derivations

A.1 Derivation of the Point-Mass Equations of Motion

Starting from Newton's second law, neglecting lift (valid for small yaw):

$$m \ddot{\mathbf{r}} = -m g \hat{\mathbf{e}}_y - \frac{1}{2} \rho C_D A \|\mathbf{v}_{\text{rel}}\| \mathbf{v}_{\text{rel}} \quad (\text{A.1})$$

Expressing in component form with $\mathbf{v}_{\text{rel}} = \mathbf{v} - \mathbf{v}_w$ and dividing by m , introducing $C_D A / (2m) = \pi C_D^{\text{std}} / (8 B_C)$, recovers Eqs. (5.10)–(5.12).

A.2 Derivation of the Coriolis Terms

In a rotating reference frame:

$$m \mathbf{a}_{\text{rot}} = \mathbf{F} - 2m (\boldsymbol{\omega} \times \mathbf{v}_{\text{rot}}) - m \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) \quad (\text{A.2})$$

The centrifugal term is absorbed into local g . In local coordinates at latitude ϕ :

$$\boldsymbol{\omega} = \omega_E (0, \cos \phi, \sin \phi)^T \quad (\text{A.3})$$

Computing the cross product, the dominant terms for $v_x \gg v_y, v_z$:

$$a_z^{\text{Cor}} \approx 2\omega_E v_x \cos \phi \quad (\text{A.4})$$

$$a_y^{\text{Cor}} \approx -2\omega_E v_x \sin \phi \quad (\text{A.5})$$

A.3 Derivation of Spin Drift via Yaw of Repose

The equilibrium yaw angle from gyroscopic precession:

$$\alpha_{\text{repose}} = \frac{g}{v^2} \cdot \frac{I_x \omega_s}{C_{M_\alpha} \frac{1}{2} \rho v^2 A d} \quad (\text{A.6})$$

This produces $F_{\text{side}} = C_{L_\alpha} \frac{1}{2} \rho v^2 A \alpha_{\text{repose}}$, causing the spin drift deflection.

Appendix B

Companion Julia Package

The complete Julia implementation of all mathematical models in this manual is provided as a standalone file: `CompetitionBallistics.jl`. This file contains six sub-modules:

1. `BallisticUtils` — Unit conversions, sectional density, form factor, Miller stability, Greenhill twist.
2. `Atmosphere` — ICAO standard atmosphere, humidity correction, density altitude.
3. `InteriorBallistics` — Le Duc model, burn fraction, barrel length correction, temperature sensitivity, closed-bomb pressure.
4. `DragModels` — G1 and G7 BRL tabular data with linear interpolation.
5. `ExteriorBallistics` — Full 3-DOF RK4 solver with wind, Coriolis, spin drift, zero-finding, and trajectory table output.
6. `ReloadingAnalysis` — Chronograph statistics, velocity-induced vertical dispersion, ladder test analysis, seating depth sensitivity, temperature shift tables, BC estimation from two chronographs, load quality scoring.

```
1 include("CompetitionBallistics.jl")
2 using .CompetitionBallistics
3 using .CompetitionBallistics.ExteriorBallistics
4 using .CompetitionBallistics.ReloadAnalysis
5
6 params = ShotParameters(
7     mass_grains=140.0, caliber_in=0.264,
8     bc=0.311, drag_model=:G7,
9     muzzle_vel_fps=2700.0, zero_range_yd=100.0,
10    wind_speed_mph=10.0, wind_angle_deg=90.0,
11    target_range_yd=1000.0,
12    twist_in=8.0, bullet_length_in=1.34,
13 )
14 trajectory_table(params, step_yd=100)
15
16 # Chronograph analysis
17 v = [2695, 2702, 2688, 2710, 2699, 2693, 2701, 2707]
18 stats = chronograph_stats(v)
19 vert = velocity_vertical_dispersion(params, stats.sd,
20                                     range_yd=1000.0)
```

Listing B.1: Quick-start usage of `CompetitionBallistics.jl`

The file requires only the Julia standard library (`Statistics`, `Printf`).

Appendix C

Greenhill's Formula and Twist Rate Selection

$$T = \frac{C d^2}{l} \sqrt{\frac{\rho_b}{\rho_a}} \quad (\text{C.1})$$

For lead-core jacketed bullets, the simplified form: $T \approx 150 d^2/l$.

Use $C = 150$ for $v < 2800$ fps, $C = 180$ for $v > 2800$ fps [[Greenhill, 1879](#)].

Stability guidance: $S_g < 1.0$ (unstable), 1.0–1.3 (marginal), 1.3–2.0 (optimal), > 2.0 (over-stabilized).

Bibliography

- Bashforth, F. (1870). *A Mathematical Treatise on the Motion of Projectiles*. Asher & Co., London.
- Carlucci, D.E. and Jacobson, S.S. (2007). *Ballistics: Theory and Design of Guns and Ammunition*. CRC Press.
- Greenhill, A.G. (1879). On the rotation required for the stability of an elongated projectile. *Minutes of the Proceedings of the Royal Artillery Institution*, 10:577–593.
- International Civil Aviation Organization (1993). *Manual of the ICAO Standard Atmosphere*. ICAO Doc 7488/3, 3rd edition.
- Ingalls, J.M. (1893). *Exterior Ballistics in the Plane of Fire*. D. Van Nostrand Company.
- Litz, B. (2011). *Applied Ballistics for Long Range Shooting*. Applied Ballistics, LLC, 2nd edition.
- Litz, B. (2015). *Ballistic Performance of Rifle Bullets*. Applied Ballistics, LLC, 3rd edition.
- McCoy, R.L. (1999). *Modern Exterior Ballistics*. Schiffer Publishing, 2nd edition.
- Miller, D. (2005). The new twist rule. *Precision Shooting*, March 2005.
- Press, W.H. et al. (2007). *Numerical Recipes*. Cambridge University Press, 3rd edition.
- Siacci, F. (1888). *Balistica*. Arnaldo Forni Editore, Bologna.

Glossary

BC	Ballistic Coefficient — ratio of sectional density to form factor.
BRL	Ballistic Research Laboratory (US Army, Aberdeen Proving Ground).
COAL	Cartridge Overall Length.
BTO	Base To Ogive length.
DA	Density Altitude.
ELR	Extreme Long Range (> 1500 yd).
ES	Extreme Spread (max velocity – min velocity).
MOA	Minute of Angle (≈ 1.047 in. at 100 yd).
Mil	Milliradian (≈ 3.6 in. at 100 yd).
OCW	Optimal Charge Weight.
PRS	Precision Rifle Series.
NRL	National Rifle League.
SD	Sectional Density or Standard Deviation (context-dependent).
VLD	Very Low Drag (bullet design).
S_g	Gyroscopic Stability Factor.
G1/G7	Standard drag model designations.